

Makkena Abhinay

☎ +91 8096248169 ✉ abhimakkena1937@gmail.com 🔗 LinkedIn 🐙 GitHub

Education

B. Tech in Artificial Intelligence and Machine Learning	2022 – 2026
Vishnu Institute of Technology, Bhimavaram	81.03%
Intermediate (MPC) — VIBM Junior College, Vijayawada	96.9%
SSC — Bhashyam High School, Addanki	99%

Technical Skills

- **Programming:** Python
- **Generative AI:** LLMs, RAG, Prompt Engineering
- **Agentic AI Frameworks:** LangChain, LangGraph, MCP
- **Machine Learning:** Scikit-learn, Computer Vision, OpenCV, NumPy, Pandas
- **Backend & APIs:** FastAPI
- **Databases:** SQL, Supabase, MongoDB
- **Web Development:** HTML, CSS, JavaScript
- **Developer Tools:** Git, GitHub, Docker

Internship Experience

AI-ML Virtual Internship (AICTE)	Jul 2025 – Sep 2025
- Gained hands-on experience with machine learning fundamentals, model building, and practical ML workflows through structured labs and assignments. - Applied machine learning techniques for data preprocessing, model training, and model evaluation. - Developed a practical understanding of supervised and unsupervised learning techniques.	

Projects

Agentic AI MCP Chatbot with RAG

- Built an agentic AI chatbot using LangGraph and Gemini with autonomous tool selection and execution across multiple tools.
- Implemented RAG for uploaded PDFs using PyPDFLoader, document chunking, Gemini embeddings, FAISS, and similarity-based retrieval.
- Integrated multiple MCP servers using MultiServerMCPClient to dynamically connect external tools and services to the agent.
- Implemented persistent conversation memory and long-term user memory using PostgreSQL, along with automatic context summarization for long conversations.

AI Daily Newsletter Assistant

- Built an agentic AI newsletter system using LangGraph and Gemini that autonomously researches and summarizes the latest AI news, startups, people, GitHub repositories, and research papers.
- Designed a multi-agent workflow with specialized Planner, Discovery, Research, Reducer, Image, Writer, Exporter, and Email agents for end-to-end newsletter generation.
- Integrated Exa for time-windowed web research and Gemini for content extraction, ranking, summarization, image selection, and newsletter generation.
- Implemented PostgreSQL-based subscriber management with FastAPI subscribe/unsubscribe APIs and automated daily newsletter delivery using GitHub Actions.

AI Educational Voice Agent

- Built an AI-powered education voice agent using FastAPI, Twilio, Gemini, and Streamlit to automatically call and qualify student leads.
- Implemented Gemini-based conversational intelligence to collect education, career goals, course interests, preferred mode, batch, and budget through voice conversations.
- Developed a Streamlit dashboard to track leads, call status, transcripts, and student information using SQLAlchemy.